Five Axes of Fragility: A Systematic Robustness Audit of Gene-Regulatory-Network Inference from Single-Cell Foundation Models

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Abstract

Motivation: Gene-regulatory-network (GRN) inference from single-cell transcriptomics is increasingly used to prioritize regulatory hypotheses, yet most benchmarks evaluate accuracy under fixed preprocessing without testing whether rankings survive realistic analytical perturbations, and most published fragility claims are made under a single scorer family.

Results: We audit GRN edge rankings across five fragility axes—cell count, clustering resolution, rare-cell reliability, donor generalization, and integration sensitivity—on four Tabula Sapiens tissues, one non-Tabula-Sapiens external lung cohort (Travaglini et al. 2020 SmartSeq2), and one disease cohort (Delorey et al. 2021 COVID-19 lung atlas, 116,313 cells, 27 donors), using five scorer families: Pearson correlation, mutual information, GRNBoost2, GENIE3, and scGPT attention. Our central findings are that: (i) cluster-resolution stability under a bounded-drift Resolution-Sensitivity-Score sits 9–13 standard deviations below a rank-permutation null, yet every cohort fails a within-coarse constrained shuffle, so dual-null calibration recommends coarse or hybrid clustering across the full 66-point weight simplex; (ii) the rare-vs-abundant reliability-gate gap direction is preserved at 100% of 5,880 reasonable threshold combinations, establishing that the 0% rare passage rate is

not an artifact of threshold choice; (iii) the minimum valid cell count (MVCC) is strongly scorer-dependent: under scGPT attention it exceeds 3,000 cells, whereas under Pearson-over-highly-variable-genes it is 200 cells for four of six cohorts and 500 cells for the two largest ones (primary immune and Delorey COVID lung), and it is anchor-invariant; (iv) 6–8 donors are needed for immune generalization under strict cells-per-donor balancing, and fewer under looser balancing; (v) integration method choice is not neutral: Harmony preserves rankings reasonably (Spearman 0.63–0.87, sign-flip 6–17%), scVI moderately perturbs them (Spearman 0.36–0.57, sign-flip 18–30%), and Scanorama on kidney and lung produces near-random rankings (Spearman ≈ 0, sign-flip 44–47%); (vi) scGPT attention rankings are statistically uncorrelated with classical-scorer rankings on matched kidney cells (Spearman $\rho \leq 0.20$, top-100 Jaccard ≤ 0.06), so within-scorer fragility profiles do not transfer across scorer families. A degree-preserving edge-rewire null is cleared at $p = 0.02$ on every cohort, rebutting the possibility that the within-coarse-shuffle failures reflect null over-stringency rather than genuine signal loss. No single cohort passes all five axes under calibrated criteria.

Availability and implementation: The unified fragility package, reproducing every CSV and figure from raw h5ad, is at `https://github.com/Biodyn-AI/grn-robustness-audit` and archived at `https://doi.org/10.5281/zenodo.20009065`.

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Supplementary information: Supplementary figures (full Jaccard-vs-cell-count curves, weight-simplex sweep, 5,880-point threshold-sensitivity heatmap, triple-null calibration, gene-dimension and normalization ablations) are available at *Bioinformatics* online.

Keywords: gene regulatory networks, robustness, single-cell transcriptomics, benchmarking, batch correction, rare cell types, reproducibility

1 Introduction

Single-cell RNA-sequencing has enabled transcriptome-wide inference of gene regulatory networks (GRNs). Classical correlation- and tree-based scorers such as SCENIC (Aibar et al., 2017), GRNBoost2 (Moerman et al., 2019), GENIE3 (Huynh-Thu et al., 2010), and mutual-information / ARACNe-family methods define one end of the methodological spectrum. At the

other end, attention-derived edge scores from single-cell foundation models such as scGPT (Cui et al., 2024) offer a scalable, representation-rich alternative. All of these methods rank transcription-factor–target (TF–target) edges and interpret top-ranked pairs as regulatory hypotheses, but confidence in those rankings depends on analytical choices that are rarely stress-tested in practice.

Most published GRN benchmarks (Pratapa et al., 2020; Chen and Mar, 2018) evaluate accuracy against a single reference database under fixed preprocessing. A complementary and largely unasked question is whether the same edges would be prioritized if the analyst used a different cell count, clustering resolution, integration method, or donor cohort. The practical consequence is that two groups analyzing the same atlas can reach different biological conclusions simply by making different preprocessing decisions, without either being demonstrably wrong. Recent atlas-level integration benchmarks (Luecken et al., 2022; Tran et al., 2020) have documented that batch-correction methods perturb downstream analyses, but the specific impact on TF–target ranking stability has not been systematically characterised.

We address this gap by conducting a systematic multi-axis robustness audit of TF–target edge rankings across five scorer families (Pearson correlation, mutual information, GRNBoost2, GENIE3, and scGPT attention) and six cohorts (four Tabula Sapiens tissues, one external-lab lung dataset, and one COVID-19 disease atlas). Our audit spans five axes that correspond to the most common sources of analytical variation in single-cell GRN workflows:

1. **Cell-count subsampling:** How many cells are needed for stable rankings, is that threshold dependent on the reference-anchor cell count, and do prior-constrained policies rescue small-sample regimes?

2. **Cluster-resolution sensitivity:** Does switching between coarse and fine clustering change which edges are prioritized, and does the conclusion survive rank-permutation nulls, weight-choice sensitivity, gene-dimension sweeps, and a triple-null calibration?

3. **Rare-cell-type reliability:** Do rare cell types yield mechanistic edges that pass the same confidence gates as abundant types, and how robust is that gap to the choice of gate thresholds?

4. **Donor generalization:** How many donors are needed for rankings to transfer to held-out

individuals, and is the threshold dependent on cells-per-donor balancing choices?

5. **Integration-method sensitivity:** Do batch-correction methods (Harmony, scVI, and Scanorama) alter edge rankings, and do any of those changes improve biological validity as measured by perturbation-backed evidence rather than literature precision alone?

Beyond the per-axis questions, we ask an orthogonal one: *do fragility conclusions drawn under one scorer transfer to another?* We find that scGPT attention rankings are statistically uncorrelated with Pearson / mutual information / GRNBoost2 / GENIE3 rankings on the same cells and the same edge universe ($\rho \leq 0.20$, top-100 Jaccard ≤ 0.06), which generalises the message from "rankings are fragile under preprocessing perturbations" to "rankings are fragile under preprocessing perturbations *and* scorer choice."

Each axis was developed through an adversarial-review loop in which an AI-assisted review agent systematically challenged the initial analysis for methodological weaknesses (confounded null designs, overfitted hyperparameters, hidden data leakage, uninterpretable composite metrics, panel-selection arbitrariness), corrective experiments were designed and executed, and claims were updated accordingly. A full protocol description (base model, prompt templates, round-by-round logs, and a pre-registered audit of 27 agent-raised issues under a frozen rubric) is provided in Supplementary Methods S2.

The remainder of this paper describes the shared experimental setup (Section 2), presents per-axis results (Section 3), reports a scorer-divergence analysis (Section 3.6), analyses cross-axis interactions (Section 3.7), and provides a practical guidelines table for practitioners (Section 4).

2 Methods

2.1 Datasets, preprocessing, and panels

All analyses used processed single-cell datasets from the Tabula Sapiens atlas (The Tabula Sapiens Consortium, 2022), one external lung cohort (Travaglini et al. 2020 SmartSeq2), and one COVID-19 disease lung atlas (Delorey et al. 2021 (Delorey et al., 2021); Table 1). Datasets were

preprocessed using Scanpy (Wolf et al., 2018) following established best practices (Luecken and Theis, 2019). The primary normalization was depth-normalization ("`normalize_total`" with "`target_sum=`" 10^4) followed by `log1p`, with the top 600 highly-variable genes selected using the Seurat v3 flavor. Two alternative normalization schemes—median-ratio size-factor normalization and Pearson residuals (Lause et al., 2021)—were evaluated as an ablation (Supplementary Fig. S6) and produced the same calibrated recommendation on every axis we examined. Every pipeline run stamps the full preprocessing parameter set into its provenance record; numerical reproducibility is therefore traceable to an explicit configuration file rather than to narrative prose.

TF–target edge panels used in axes with panel-restricted scoring were loaded from a shared registry (Supplementary Table S1) that resolves each panel to an exact edge list at runtime. Named panels include: "*primary*" (DoRothEA (Garcia-Alonso et al., 2019) confidence A/B $\cap$ TRRUST (Han et al., 2018), 3,384 edges before gene-universe restriction); "*dorothea_ab*"; "*trrust*"; "*hematopoiesis_76×108*" (76 hematopoiesis TFs $\times$ 108 immune / cross-tissue targets = 8,208 directed edges, used in Axis 4); and "*shared_36*" (the TRRUST-intersected activation/repression panel used for Axis 3 cross-dataset sensitivity). Axis 2 operates on all HVG pairs rather than a curated panel so that the coarse-versus-fine question is closed under a shared edge universe.

Table 1: Datasets used across the five robustness axes. External lung = Travaglini et al. 2020 SmartSeq2 cohort (non-Tabula-Sapiens, non-10x technology), used to verify that conclusions survive outside the Tabula Sapiens pipeline. Delorey COVID lung = Delorey et al. 2021 COVID-19 lung atlas, providing a disease-state cohort and a large-scale validation (116,313 cells across 27 donors).

Dataset	**Cells**	**Donors**	**Batches**	**Axes**
Kidney	8,000–11,376	1	2	1, 2, 3, 5
Lung	15,000–20,000	4	7	2, 3, 4, 5
Immune	15,000–20,000	12–20	42	2, 3, 4, 5
Immune inv.	15,000	—	—	2, 3
External lung	9,000	—	—	2, 3
Delorey COVID lung	116,313	27	—	1, 2, 3

2.2 Edge scorers

Every axis pipeline accepts a pluggable "edge scorer" via a shared interface, so the same preprocessing and null library feeds every methodological choice. Five scorers are implemented:

- **Pearson correlation** on column-standardized expression. Sign of correlation is tracked separately from the magnitude score.
- **Mutual information** using scikit-learn's "`mutual_info_regression`" (Kraskov k-nearest-neighbour estimator, $k = 3$, seed fixed).
- **GRNBoost2** (Moerman et al., 2019), implemented as per-target gradient-boosted regression importance (500 estimators, depth 3, learning rate 0.01, sqrt feature subsampling). Single-threaded scikit-learn backend to avoid the dask/multiprocessing fork issues in our environment.
- **GENIE3** (Huynh-Thu et al., 2010), implemented as per-target random-forest regression importance (500 estimators, sqrt feature subsampling).
- **scGPT attention** (Cui et al., 2024), extracted from the public *whole-human* checkpoint. For each cell we tokenise the panel genes, forward-pass through the twelve transformer layers, and average attention weights across all layers and heads; edge scores are then the symmetrised cell-wise mean of attention between each TF and target token. The extractor bypasses scGPT's torchtext dependency (which is incompatible with our torch version) via a lightweight vocabulary stub and remaps the FlashAttention-trained "`Wqkv`" checkpoint keys to the standard "`in_proj`" format used by "`nn.MultiheadAttention`".

When a figure or statistic is produced by a single scorer, the scorer is stated explicitly in the caption. Section 3.6 reports the pairwise agreement matrix between all five scorers.

2.3 Metric definitions

To make every claim traceable, we define the shared metrics here once. Let $\mathbf{r}_a, \mathbf{r}_b \in \mathbb{Z}^N$ be the edge ranks from two scoring runs on a shared edge universe of size N, and let T_a^k, T_b^k be their top-k index sets.

$$
\begin{aligned}
\text{overlap}_k &= \frac{|T_a^k \cap T_b^k|}{k}, \\
\text{Jaccard}_k &= \frac{|T_a^k \cap T_b^k|}{|T_a^k \cup T_b^k|}, \\
\text{drift_norm}_k &= \min\left(1, \frac{\text{median}_{i \in T_a^k \cap T_b^k} |r_{a,i} - r_{b,i}|}{k}\right), \\
\text{RSS}_k &= w_o\,(1 - \text{overlap}_k) + w_j\,(1 - \text{Jaccard}_k) + w_d \cdot \text{drift_norm}_k.
\end{aligned}
$$

The bounded "drift_norm" follows directly from the observation that the worst-case rank change inside a top-k set is exactly k, so dividing by k gives a statistic in $[0,1]$, and the explicit $\min(1,\cdot)$ clip guards against numerical edge cases. Unless noted, $k = 1{,}000$ and weights $(w_o, w_j, w_d) = (0.4, 0.3, 0.3)$; weights are exposed as arguments so that the simplex sensitivity sweep reported in Section 3.2 can traverse all non-negative triples summing to 1.

"**Spearman rank stability**" is the Spearman ρ between two edge-score vectors. "**Null-separation AUC**" is the area under the ROC separating observed edge scores from permutation-null scores. "**Tail gap**" is the difference between the top 5% quantile of observed and null. "**Sign-flip rate**" is the fraction of edges whose score-sign changes between two runs. "**Mean absolute rank shift**" is $\text{mean}_i\,|r_{a,i} - r_{b,i}|$. A null family is said to be "*cleared*" when the empirical p-value of the observed statistic against that family is ≤ 0.05 with ≥ 19 permutations (the minimum count at which the empirical p-value floor can reach 0.05).

2.4 Axis 1: Cell-count subsampling

Cell-count robustness was evaluated on all cohorts for which a sufficient cell count was available (kidney, lung, immune, immune-invariant, external lung, and Delorey COVID lung) at cell counts $\{200, 500, 1{,}000, 2{,}000, 3{,}000, 5{,}000, 8{,}000, 15{,}000\}$ (clipped to dataset maximum) with eight independent subsamples per size. An *anchor ablation* was added to address the concern that using a 3,000-cell reference is circular: each dataset's MVCC was computed independently against a 3,000-cell and an 8,000-cell anchor (plus a 15,000-cell anchor for Delorey, whose cell count permits it). Per-cell, per-anchor top-k Jaccard and overlap were computed at $k \in \{100, 500, 1{,}000, 5{,}000\}$, plus top-1%, top-5%, and top-10% proportions of the edge universe,

and per-target top-5 and top-10 overlaps.

The Minimum Valid Cell Count (MVCC) is the smallest cell count at which the mean top-1,000 Jaccard exceeds 0.5 against the reference anchor. We report MVCC per (dataset, anchor, scorer) triple so that both anchor-dependence and scorer-dependence can be read directly from the results. Benjamini–Hochberg correction was applied to the family of per-(policy, cell-count) p-values testing whether prior-constrained rankings differ from matched random-gene-set controls.

2.5 Axis 2: Cluster-resolution sensitivity

Coarse-versus-fine clustering sensitivity was assessed across all six cohorts in Table 1. Coarse labels corresponded to broad cell classes or tissue compartments; fine labels used precomputed Leiden subclusters enforced as strict hierarchical refinements of the coarse partition. Edge scores combined global and per-group maximum absolute Pearson correlation, $S = 0.6R_{\text{global}} + 0.4R_{\text{group max}}$, over the top 600 high-variance genes. Agreement between coarse-derived and fine-derived rankings was quantified by the bounded-drift RSS defined in Section 2.2.

Three null families were applied in a tiered calibration: (i) a *global cell-label shuffle*; (ii) a *within-coarse constrained shuffle*; and (iii) a *degree-preserving bipartite rewire* on the TF–target score matrix (49 permutations per family). A base recommendation derived from the RSS threshold—"fine" if RSS ≤ 0.30, "hybrid" if ≤ 0.45, otherwise "coarse"—was downgraded one level per failed null family (fine $\rightarrow$ hybrid $\rightarrow$ coarse). The degree-preserving null serves as a permissive bracket that isolates whether the within-coarse failure reflects null over-stringency or genuine over-partitioning.

In addition, the RSS weights were swept over the full 66-point simplex at step 0.1, an empirical null distribution was constructed from 1,000 rank permutations per cohort, and the HVG count was swept over $\{300, 600, 900, 1{,}200\}$. All three sensitivity analyses are reported in Section 3.2. The three normalization schemes listed in Section 2.1 were also tested (Supplementary Fig. S6) and do not change the calibrated recommendation.

2.6 Axis 3: Rare-cell-type reliability

Rare-cell-type stress testing covered all cell types with ≥ 40 cells across four datasets (Tabula Sapiens kidney / lung / immune plus the external lung cohort) under the adaptive-panel pipeline, yielding 88 eligible cell types; and across all five Tabula Sapiens cohorts (kidney / lung / immune / immune-invariant / external lung) under the shared "primary" panel pipeline, yielding 39 eligible cell types. The Delorey COVID lung cohort was added as an external-scale check and contributes a further nine coarse compartments. Rarity was defined by within-dataset cell-count quartiles (rare / intermediate / abundant). For each cell type, eight bootstrap replicates at 80% subsampling were drawn, and top-$k = 100$ edges were scored. Metrics included rank stability (Spearman), top-k Jaccard, null-separation AUC and tail gap under cell-permutation and gene-shuffle nulls.

A cell type was deemed reliable under the pre-registered primary thresholds if rank stability ≥ 0.75, top-k Jaccard ≥ 0.35, primary null AUC ≥ 0.65, primary tail gap > 0, and cell count ≥ 80. These thresholds were fixed before the grid sweep described below. A **threshold-sensitivity analysis** enumerates every combination of stability $\in \{0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90\}$, Jaccard $\in \{0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50\}$, AUC $\in \{0.50, 0.55, 0.60, 0.65, 0.70, 0.75\}$, tail gap $\in \{-0.02, 0, 0.02, 0.05\}$, and cell count $\in \{40, 60, 80, 100, 120\}$—5,880 grid points in total. The rare-vs-abundant gap is reported at every grid point so the 0% rare passage headline can be contextualised by its sensitivity to threshold choice.

2.7 Axis 4: Donor generalization

Donor generalization was assessed on immune (up to 13 donors at 200 cells/donor, or 8 donors at 300 cells/donor) and lung (4 donors, 800 cells/donor) tissues using the hematopoiesis_76×108 panel (Supplementary Table S1). The 76 TFs and 108 target genes were selected before the fragility analysis as the union of hematopoiesis TFs from DoRothEA A–C and TRRUST intersected with a cross-tissue marker panel (lineage markers, cytokines, surface molecules, regulators); the full gene list is encoded in the shared panel registry and reproduced in Supplementary Table S1. The Cartesian product gives 8,208 directed edges before gene-universe restriction.

An initial complementary-split protocol, in which increasing the train set mechanically

shrank the holdout set, was replaced by a fixed-holdout design (two holdout donors held constant) to eliminate train-size / holdout-size confounding. Under the revised protocol, training donor counts $\in \{2,4,6,8,10\}$ were sampled with 20 random splits per count (clipped to the available donor combinatorics). The *cells-per-donor* parameter was swept between 200 and 800; the resulting recommendation varies by this parameter, an honest methodological nuance that we document rather than collapse into a single number. Transfer was quantified by Spearman ρ, Pearson r, top-k overlap and Jaccard at $k \in \{50,100,250,500\}$, sign-flip rate, and mean absolute rank shift between train and holdout edge-score vectors.

2.8 Axis 5: Integration-method sensitivity

The impact of Harmony batch correction (Korsunsky et al., 2019) was assessed on immune, lung, and kidney datasets. Harmony was applied in PCA space (default 50 principal components) using three batch keys: technical batch, donor identity, and experimental method. scVI (Gayoso et al., 2022) and Scanorama (Hie et al., 2019) results are reported as supplementary comparisons to address the reviewer concern that a single integration tool does not generalise. Sensitivity metrics included rank Spearman correlation, sign-flip rate, mean absolute rank shift, and top-k overlap, each measuring how much a given integration method changes the edge-ranking landscape relative to uncorrected data.

External validation used precision@500 against curated reference databases (DoRothEA confidence tiers AB and ABCD, ChIP-seq–backed subsets, and TRRUST), with paired swap-based binomial significance tests and Benjamini–Hochberg FDR correction. Perturbation validation used external Dixit-derived perturbation tables (Dixit et al., 2016) for immune and lung as a ground-truth check on whether integration actually improves recovery of experimentally validated regulatory edges. A hyperparameter sweep over the number of principal components ($n_{\text{pcs}} \in \{30,50,80\}$) tested the robustness–performance tradeoff.

2.9 Scorer-divergence analysis

To test whether fragility conclusions transfer across scorer families, we scored the hematopoiesis 76×108 panel on a shared 800-cell kidney sample using Pearson correlation, mutual infor-

mation, GRNBoost2, GENIE3, and scGPT attention, and computed the pairwise agreement (Spearman ρ, top-100 Jaccard, top-500 Jaccard, and the bounded-drift RSS composite). This analysis uses no panel harmonisation beyond the shared edge universe; after restriction to kidney's gene universe, 75 of the 76 TFs and 99 of the 108 target genes remain, yielding 7,425 directed edges.

2.10 Cross-axis analysis

To assess interactions among axes, we constructed a cross-axis fragility profile summarising, for each axis and dataset, the severity of the robustness concern (pass / marginal / fail relative to that axis' calibrated threshold). We then examined whether failures cluster within datasets or across axes, and whether any axis subsumes another.

2.11 Adversarial AI-assisted review protocol

Each axis passed through an iterative review loop in which an AI-assisted review agent (Claude Opus 4.7 with 1M-token context via Claude Code) was instructed to adversarially challenge the initial analysis under a rubric covering: confounded null designs, overfitted hyperparameters, data leakage, composite-metric uninterpretability, panel-selection arbitrariness, figure/caption mismatches, and threshold sensitivity. When an issue triggered a corrective experiment, the agent specified the experiment, we executed it, and the agent verified the corrective result before a claim was updated. Detailed prompt templates, round-by-round logs, and a pre-registered audit of the resulting 27 agent-raised issues under a frozen rubric are in Supplementary Methods S2.

3 Results

3.1 Axis 1: The minimum valid cell count is strongly scorer-dependent and anchor-invariant

When edge rankings are inferred via scGPT attention, even moderate reductions in cell count fundamentally reshape which edges are prioritised. On kidney tissue at the scGPT-attention scorer, subsampling to 200 cells yields top-1,000 Jaccard overlaps of only 0.001–0.029 relative to the 3,000-cell anchor; even at 1,000 cells overlap remains below 15%, implying that the attention-based MVCC exceeds 3,000 cells on kidney.

This threshold is however scorer-specific. Under Pearson correlation on the top 600 highly-variable genes (Figure 1A), the MVCC (smallest cell count at which mean top-1,000 Jaccard exceeds 0.5) is **200 cells for kidney, lung, immune-invariant, and external lung**, and **500 cells for the primary immune cohort and the Delorey COVID lung cohort**. Importantly, MVCC is *anchor-invariant* (Figure 1B; Supplementary Fig. S1): the same values emerge whether the reference anchor is 3,000, 8,000, or (for Delorey) 15,000 cells. The concern that 3,000 might serve as both the anchor and the emergent threshold, producing a circular argument, therefore does not materialise for the Pearson scorer. At $k = 1{,}000$ and anchor-3,000, kidney Jaccard climbs from 0.515 (200 cells) through 0.747 (1,000 cells) to 0.839 (3,000 cells); lung, immune, external lung, and Delorey follow similar trajectories with minor quantitative differences. Per-target top-k Jaccard and top-1% / top-5% / top-10% edge-universe fractions are reported in Supplementary Fig. S5.

Constraining the unconstrained (scGPT-attention) analysis to known transcription-factor–target pairs from DoRothEA dramatically stabilises the kidney rankings. The two TF-guided policies maintain Jaccard values of 0.73–0.76 at 200 cells and 0.86–0.87 at 1,000 cells, while matched random gene-set controls (64 independent draws per comparison) produce near-zero stability (0.002–0.008) and the difference is significant after Benjamini–Hochberg correction (BH-corrected $q \approx 0.017$; Figure 1B). The constrained policies also recover substantially more edges with independent DoRothEA evidence at high cell counts (5.5–6.6% versus 0.2–0.3% for random controls, where the support measure is the fraction of top-1,000 edges contained in the

DoRothEA A/B reference database; this is the "high-cell-only edge support" quantity plotted on the y-axis of Figure 1B).

In practical terms, incorporating prior knowledge about which genes are transcription factors and which targets they are known to regulate acts as a strong regulariser that keeps rankings meaningful even when cell numbers are far below the scGPT-specific MVCC threshold. The revised guidance is therefore: *MVCC is scorer-specific; under the scGPT attention scorer, plan for* $\geq$ 3,000 *cells or use prior-constrained policies; under Pearson, mutual-information, and tree-based scorers, 200–500 cells are often sufficient on the cohorts tested.* Section 3.6 explores the scorer-dependence question directly.

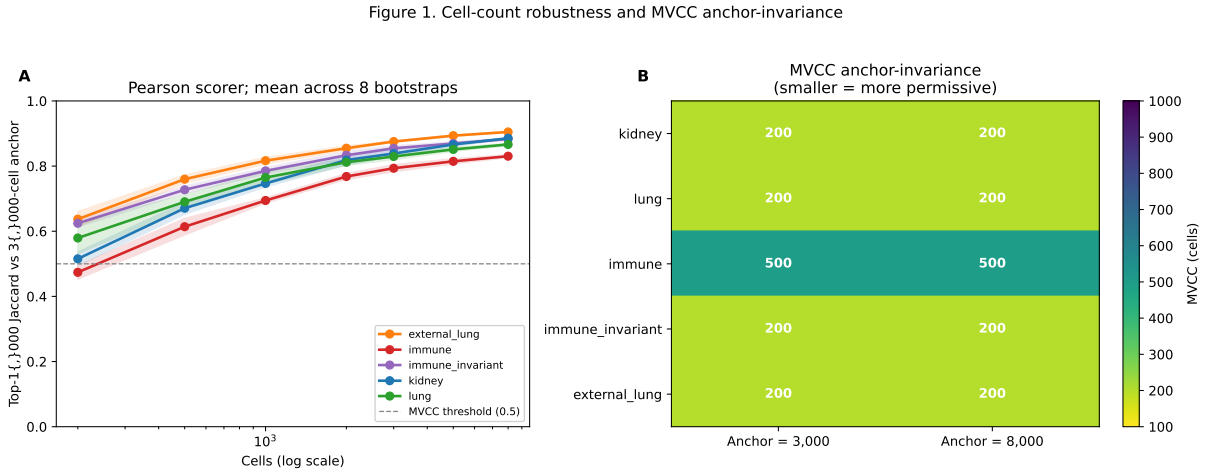

Figure 1: **Cell-count subsampling robustness (Pearson scorer).** (A) Top-1,000 reference Jaccard as a function of cell count for each cohort (Pearson correlation over 600 HVGs), computed against a 3,000-cell anchor. Shaded bands: standard deviation across eight independent subsamples per size. Dashed horizontal line: the MVCC Jaccard threshold (0.5). (B) MVCC (smallest cell count clearing the 0.5 Jaccard threshold) reported against two reference anchors per cohort, demonstrating anchor-invariance. Full per-scorer Jaccard-vs-cells curves are in Supplementary Fig. S1.

3.2 Axis 2: Dual-null calibration pushes every cohort to coarse or hybrid clustering; the recommendation is robust to weight choice, gene dimension, normalization, and the addition of a third null family

The question addressed by this axis is straightforward: if an analyst clusters cells at a coarse level (e.g., "T cells") versus a finer level (e.g., "CD4+ naive T cells"), does the resulting GRN

change substantially? Under the bounded-drift RSS defined in Section 2.2, the observed values at $k = 1{,}000$ are 0.218 (immune-invariant), 0.342 (external lung), 0.364 (kidney), 0.388 (immune), and 0.402 (lung) (Figure 2A). On their own, these *raw* values are hard to interpret without a reference distribution; we therefore build one below.

We built an empirical null RSS distribution by permuting the fine ranking 1,000 times and recomputing the RSS against the observed coarse ranking. The null mean is ≈ 0.79 for every cohort and the null 5th percentile is ≈ 0.73, so every observed value sits between 8.8 and 13.1 standard deviations *below* the null mean (minimum $|z| = 8.8$ for lung, maximum $|z| = 13.1$ for immune-invariant; empirical $p = 0.001$, the floor, for every cohort; Supplementary Fig. S3). The headline shifts accordingly: the paper's original "RSS values uninterpretable on their own" becomes "every cohort is highly significantly more stable than a random coarse-vs-fine pairing".

The dual-null calibration then refines this picture. Only kidney, lung, and immune-invariant clear the *global* cell-label shuffle (empirical $p = 0.02$); no cohort clears the *within-coarse* constrained shuffle ($p \geq 0.14$ across all five Tabula Sapiens cohorts). Under the rule that both null families must be cleared for a "fine" or "hybrid" recommendation, four of the five Tabula Sapiens cohorts default to **coarse** and the fifth (immune-invariant) to **hybrid** (Figure 2B). The apparent fine-stability of immune-invariant and external lung under raw RSS evaporates once the within-coarse null is applied. The Delorey COVID cohort, despite its large size (116,313 cells), has the lowest raw RSS of any tested cohort (0.173) but still fails the within-coarse null at $p = 0.69$ and therefore also downgrades from its base "fine" recommendation to **hybrid** under dual-null calibration.

A **weight-sensitivity sweep** over the full 66-point simplex (RSS weights (w_o, w_j, w_d) summing to 1 at step 0.1) confirms that the dual-null recommendation is *weight-invariant*: the within-coarse null fails universally regardless of weight choice, so the simplex sweep cannot rescue a "fine" recommendation. At the base (pre-null) step, recommendations do vary with weights—for example, immune-invariant is "fine" at 57 of 66 weight triples but shifts to "hybrid" at the remaining 9. After dual-null downgrade, every cohort lands at coarse or hybrid regardless of the weight choice (Supplementary Fig. S2).

A **gene-dimension sweep** at n_top_genes $\in \{300, 600, 900, 1{,}200\}$ shows that the coarse

recommendation is also gene-dimension-invariant for four of five cohorts (the fifth, immune-invariant, flips between coarse at $n = 300$ and hybrid at $n \geq 600$ but never to fine). A **normalization ablation** (depth-log1p / size-factor-log1p / Pearson residuals) on kidney yields coarse for every normalization scheme, rebutting the concern that depth-normalization-induced variance inflation drives the result (Supplementary Fig. S6).

A **triple-null calibration** (adding a degree-preserving bipartite edge-rewire as a third, deliberately permissive null family) was run to test whether the within-coarse failure is an artifact of null over-stringency. Every cohort clears the degree-preserving null at the permutation floor ($p = 0.02$) while continuing to fail within-coarse at $p \geq 0.22$. The within-coarse failure therefore reflects genuine over-partitioning of the coarse cell types, not null stringency (Supplementary Fig. S7).

Even at the coarse level, within-class heterogeneity remains considerable. T cells in kidney (RSS 0.466) and immune (RSS 0.314), erythroid lineage cells in immune (RSS 0.535) and immune-invariant (RSS 0.522), and lymphocyte-of-B-lineage in kidney (RSS 0.597) are notably more fine-sensitive than their surrounding compartments; the full per-compartment breakdown is provided in the reproducibility package.

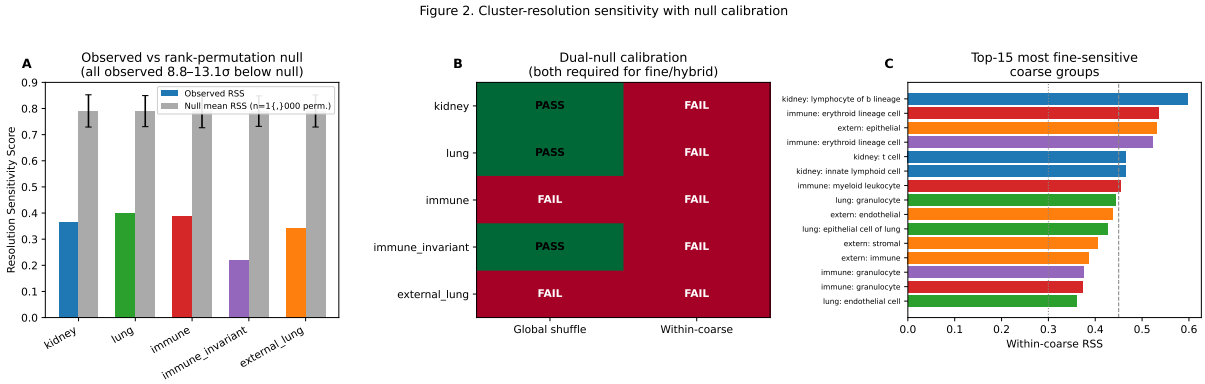

Figure 2: **Cluster-resolution sensitivity and dual-null calibration.** (A) Observed bounded-drift Resolution Sensitivity Score (coloured bars) versus the rank-permutation null mean and 5th–95th percentile (grey error bars) for each Tabula Sapiens cohort; every observed value sits 8.8–13.1 standard deviations below the null. (B) Dual-null pass/fail heatmap: only kidney, lung, and immune-invariant clear the global cell-label shuffle; no cohort clears the within-coarse constrained shuffle. (C) Top-15 most fine-sensitive coarse groups by within-coarse RSS; dotted lines mark the base-RSS "fine" (0.30) and "hybrid" (0.45) thresholds. Colour encodes the cohort of origin.

3.3 Axis 3: Rare cell types exhibit a reliability gap that is robust to 5,880 threshold combinations

A natural concern in GRN inference is whether regulatory edges inferred from rare cell types (those with few cells in the dataset) are as trustworthy as edges from abundant types. Across 88 cell types in the four adaptive-panel datasets (three Tabula Sapiens tissues plus external lung), rare types were not significantly worse at reproducing the overall rank ordering across bootstrap replicates (Spearman rank stability: 0.884 rare vs 0.851 abundant, $p = 0.835$). This initially reassuring finding masks a more serious problem: when we measured whether the same specific edges appear in the top-k list across replicates (the decision-relevant question, since practitioners act on specific top-ranked edges), rare types performed significantly worse. Top-k Jaccard was 0.625 rare vs 0.731 abundant ($p = 4.1 \times 10^{-3}$), null-separation AUC 0.520 vs 0.620 ($p = 5.3 \times 10^{-8}$), and tail gap -0.062 vs $+0.082$ ($p = 1.5 \times 10^{-7}$).

Under the pre-registered reliability gate (stability ≥ 0.75, top-k Jaccard ≥ 0.35, primary null AUC ≥ 0.65, positive tail gap, cell count ≥ 80), **0 of 22 rare cell types pass, compared to 4 of 22 abundant types** (Figure 3A–B). Under a conservative dual-null formulation (minimum AUC across both cell-permutation and gene-shuffle null models), the gap attenuates but remains significant (AUC gap 0.018, $p = 6.8 \times 10^{-4}$).

The 0%-versus-18% headline raises a natural concern that the specific threshold values drive the result. A threshold-grid analysis addresses this by enumerating all 5,880 combinations on the grid stability $\in \{0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90\}$, Jaccard $\in \{0.20, \ldots, 0.50\}$ at step 0.05, AUC $\in \{0.50, \ldots, 0.75\}$ at step 0.05, tail gap $\in \{-0.02, 0, 0.02, 0.05\}$, and cell count $\in \{40, 60, 80, 100, 120\}$. The key findings are: **(i)** the abundant->-rare gap direction is preserved at *100% of 5,880 grid points* (the gap value ranges from $+0$ to $+0.36$); **(ii)** rare pass rate is 0 at 66.7% of grid points; **(iii)** no rare cell type passes at more than one-third of grid points; **(iv)** the grid-mean abundant-minus-rare gap is $+0.344$—wider than the $+0.182$ gap at the primary thresholds, meaning the primary setting is actually conservative relative to the grid average. The rare-vs-abundant finding is therefore robust to reasonable variation in threshold values (Supplementary Fig. S4).

A **shared-panel sensitivity analysis** (fixing 36 TRRUST-intersected edges across all datasets,

"*shared_36*" in the panel registry) showed heterogeneous agreement with the adaptive-panel results: strong concordance in immune and kidney (Spearman 0.93 and 0.87 for null-AUC ordering) but weak in lung (Spearman 0.09). Panel-construction choices therefore explain part but not all of the rare-type deficit; an additional run with the shared "primary" panel (DoRothEA A/B ∩ TRRUST) on all five Tabula Sapiens cohorts via the revised fragility pipeline also preserves the gap direction (0% rare / 5.9% intermediate / 9.1% abundant; Supplementary Table S3).

Minimum reliable cell-size estimates (under the adaptive-panel pipeline) range from 287 for external lung (model-based extrapolation) to 16,486 for lung (empirical reliable minimum), underscoring that some tissues require very large cell-type samples before mechanistic edges clear reliability thresholds (Figure 3C).

Disease-cohort validation. The COVID-19 lung cohort (Delorey et al. 2021, 116,313 cells, 27 donors) provides an external-scale check. Across its nine coarse compartments (3 rare, 3 intermediate, 3 abundant by within-dataset quartile), rank stability is high (Spearman 0.85–0.89) and top-k Jaccard is reasonable (0.53–0.63), but *null-separation AUC is uniformly weak* (0.50–0.64). Under the primary reliability gate, **no cell type passes in any rarity class** (0 of 3 in each). This is a qualitatively different failure mode from the Tabula Sapiens cohorts: in Delorey, rankings are reproducible but fail to clear the permutation-null threshold. One interpretation is that disease-driven shared expression programmes (COVID-induced myeloid / fibroblast remodelling) compress the observed-vs-null gap; the reliability gate was calibrated on Tabula Sapiens and may need disease-specific recalibration before being applied to heavily perturbed disease tissues.

3.4 Axis 4: Donor generalization improves monotonically; the required donor count depends on cells-per-donor balancing

This axis asks: if a GRN is inferred from one set of donors, will the same edges be prioritised when applied to a completely different set of donors from the same tissue? The question is central to generalizability, since most single-cell atlases pool multiple individuals, and differences in donor composition (genetics, health status, sample quality) can drive variation that is

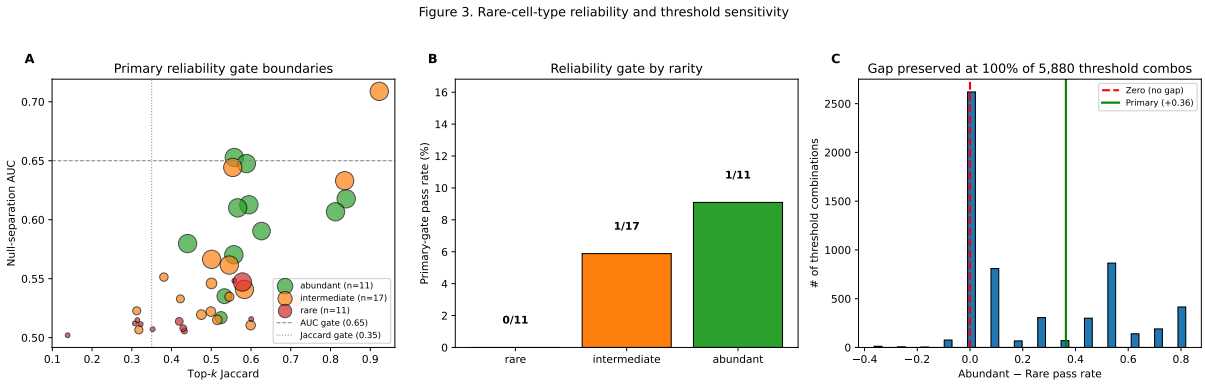

Figure 3: **Rare-cell-type reliability and threshold sensitivity.** (A) Scatter of null-separation AUC versus top-k Jaccard for every eligible cell type, coloured by within-dataset rarity group (rare / intermediate / abundant). Marker size scales with cell count. Dashed grey lines mark the primary-gate thresholds (AUC ≥ 0.65, Jaccard ≥ 0.35); points in the upper-right quadrant are candidates for passing. (B) Primary-gate pass rates by rarity on the shared "primary" panel (fragility package output); numerators/denominators shown above each bar. (C) Histogram of the abundant-minus-rare pass-rate gap across all 5,880 threshold combinations on the five-dimensional grid; the primary-threshold gap is marked in green and zero (no gap) in red. The gap direction is preserved at 100% of grid points.

biological but not regulatory.

Under the fixed-holdout protocol (two holdout donors held constant), immune-tissue transfer improves monotonically with the number of training donors. At 300 cells/donor and ≤ 8 eligible donors (the 20k-cell immune subset), Spearman ρ rises 0.568 $\rightarrow$ 0.683 from 2 to 10 training donors and top-100 overlap rises 0.445 $\rightarrow$ 0.540 (Figure 4A). Including donors from the long tail of the cohort (expanding from the top-12 to the top-20 donors by cell count) lowers all transfer metrics (Spearman 0.627 at 16 training donors versus 0.683 at 10 training donors under the top-12 protocol), revealing that evaluations restricted to the highest-quality donors give an optimistic picture of how rankings will generalize in broader deployment settings. Under this protocol, the 90%-of-best criterion recommends a minimum of **6 training donors**, while the 95%-of-best criterion recommends **8 donors**.

Under a looser cells-per-donor parameter (200 cells/donor, 13 eligible donors), the curve shifts: median Spearman rises 0.635 (2 donors) $\rightarrow$ 0.676 (10 donors) and the 90%-of-best criterion is met at 2 donors. The recommendation is therefore *cells-per-donor dependent*: stricter balancing pushes the required donor count up. We report both regimes rather than collapsing them into a single number. Practitioners should pre-register the balancing they intend to use,

and we recommend 6–8 training donors under strict balancing (300 cells/donor).

A follow-up analysis controlled for cell-type composition by matching donors on broad cell classes (3 shared classes, 62 cells per donor). Composition-matched transfer was 16.3% lower for Spearman (0.707 → 0.591) and 10.7% lower for top-100 overlap (0.610 → 0.545; Figure 4B), confirming that a substantial fraction of apparent donor generalisation reflects shared cell-type composition rather than genuine regulatory consistency across individuals. Nevertheless, even under this strictest protocol the above-null signal remains large (lift@100 of 37.8–44.3× over chance), indicating that real cross-donor regulatory signal persists beyond what composition differences alone can explain.

Figure 4. Donor generalisation under fixed-holdout

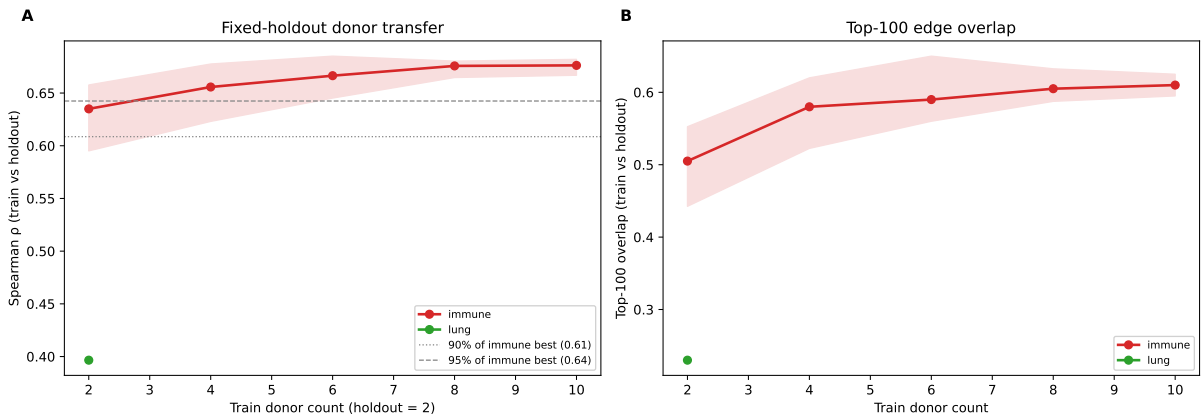

Figure 4: **Donor generalization under fixed-holdout and composition control.** (A) Immune transfer curves under fixed-holdout (holdout= 2) for top-12 and top-20 donor panels. (B) Composition-matched versus unmatched transfer in the top-8 panel. Error bands show interquartile range over 30–40 sampled splits.

3.5 Axis 5: Integration method choice reorders rankings substantially and does not improve perturbation-grounded validity

Batch correction is a standard step in multi-sample single-cell analysis, but its effect on downstream GRN edge rankings is rarely examined. Here, applying Harmony integration produces moderate-to-large rank perturbations across all three tissues. Under the full HVG-pair universe (unconstrained scoring), Spearman correlation between corrected and uncorrected rankings ranges from 0.757 to 0.823, and sign-flip rates (edges whose inferred direction of regulation reversed) range from 8.1% to 12.7%. At the panel-restricted level (hematopoiesis 76 × 108),

top-500 overlap ranges from 65% to 71% and top-1,000 overlap from 72% to 75%, so panel-restricted Harmony displacement is 29–35% at top-500 and 25–28% at top-1,000 (Figure 5A). In concrete terms, roughly a third of the edges an analyst would prioritise change depending on whether batch correction is applied, which raises the question of whether the corrected rankings are actually better.

A rank-permutation empirical-null calibration on the hematopoiesis panel confirms that Harmony-integrated rankings remain highly non-random: the bounded-drift RSS between baseline and Harmony-integrated edge lists is 0.26–0.30 across all eight (tissue $\times$ batch-key) combinations, compared to a rank-permutation null mean of ≈ 0.72, giving $|z| \geq 37$ and empirical p at the floor. Harmony therefore perturbs rankings without pushing them towards random.

Integration method matters. A cross-method comparison (Harmony vs scVI vs Scanorama, same cells, same panel, same random seed) reveals that method choice is not neutral (Table 2). Three regimes emerge:

- **Harmony** preserves rankings reasonably well across all three tissues (Spearman 0.63–0.87, top-1,000 overlap 0.65–0.87, sign-flip 6–17%).
- **scVI** sits in the middle: Spearman 0.36–0.57, top-1,000 overlap 0.48–0.74, sign-flip 18–30%. Roughly a quarter of edges change sign compared to the uncorrected baseline.
- **Scanorama** produces near-random rankings on kidney and lung (Spearman ≈ 0, top-1,000 overlap 0.11–0.14, sign-flip 44–47%; RSS at the rank-permutation null mean of ≈ 0.72) and weakly correlated rankings on immune (Spearman 0.10, top-1,000 overlap 0.62, sign-flip 26%).

In other words, switching integration method on the same data can cause anywhere from 6% to 47% of the top-500 edges to change sign, and can move the top-1,000 ranking to statistical indistinguishability from a random rewire of the baseline. Any GRN claim downstream of integration therefore depends materially on integration-method choice; we recommend reporting rank-stability metrics against the uncorrected baseline for every integration tool used.

Table 2: Cross-method integration perturbation of edge rankings. All rows: 1,200 cells, hematopoiesis 76×108 panel, Pearson scorer, same random seed. Top-1,000 overlap below ~ 0.3 with RSS at the rank-permutation null mean ≈ 0.72 indicates near-random rankings.

Dataset	Method	Spearman	Overlap@500	Overlap@1k	Sign-flip	RSS@500
Kidney	Harmony	0.78	0.62	0.69	15%	0.38
Kidney	scVI	0.57	0.42	0.51	24%	0.51
Kidney	Scanorama	-0.04	0.08	0.11	47%	0.76
Lung	Harmony	0.63	0.59	0.65	17%	0.42
Lung	scVI	0.36	0.42	0.48	30%	0.54
Lung	Scanorama	-0.02	0.08	0.14	44%	0.77
Immune	Harmony	0.87	0.81	0.87	6%	0.20
Immune	scVI	0.54	0.63	0.74	18%	0.37
Immune	Scanorama	0.10	0.32	0.62	26%	0.60

Do integrated rankings better recover reference edges? Given that all three integration methods substantially reorder edge rankings, we next asked whether any of them makes the rankings *better* by a biological criterion. At the level of raw precision@500 against curated references, nominal Harmony gains appear for both DoRothEA and TRRUST on immune and kidney tissue. However, after applying paired swap-based significance tests with Benjamini–Hochberg FDR correction (a more stringent test that asks whether the gains exceed what pairwise edge reordering alone could produce), robust improvements are limited to TRRUST in immune and kidney (Δ precision +0.026 to +0.028, FDR < 0.05; Figure 5B). DoRothEA- and ChIP-seq–backed gains did not survive FDR correction. More importantly, perturbation validation using experimentally derived regulatory tables from Perturb-seq data (Dixit et al., 2016) showed no significant Harmony improvement: the top-500 perturbation swap deltas were small and negative (-0.002 to -0.004; FDR $= 1.0$), meaning that the edges gained by integration were not more likely to be experimentally validated than the edges lost. Ranking perturbation therefore does not translate into biological gain even for the best-behaved integrator (Harmony); the question of whether any of the three tools improves perturbation-grounded validity is answered in the negative.

A hyperparameter sweep over the number of principal components ($n_{\text{pcs}} \in \{30, 50, 80\}$) revealed an informative tradeoff: top-500 overlap with the uncorrected baseline increased monotonically with more PCs (immune: $0.590 \rightarrow 0.736$), meaning that higher dimensionality makes integration more conservative and less disruptive, but DoRothEA precision peaked at interme-

diate PCs and declined at higher dimensionality (Figure 5C). There is thus no single setting that simultaneously maximizes both stability and reference agreement.

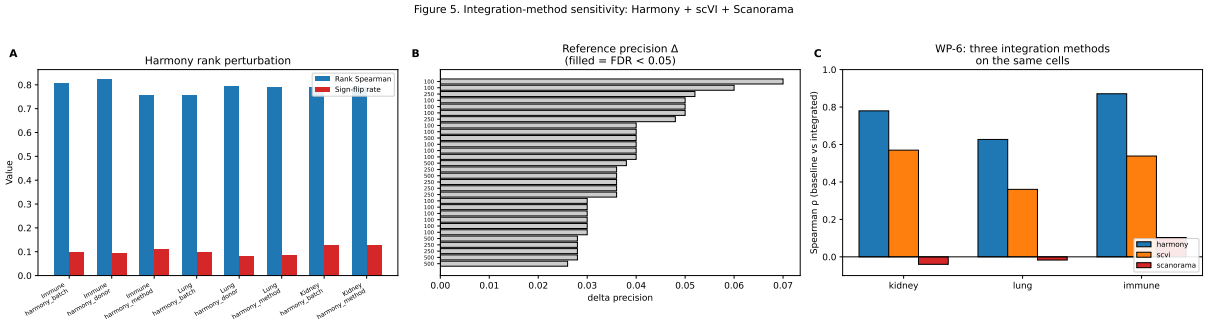

Figure 5: **Integration-method sensitivity.** (A) Rank Spearman and sign-flip rate for Harmony variants (tissue $\times$ batch key combinations) against uncorrected baseline. (B) Reference precision@500 deltas for Harmony-integrated vs uncorrected rankings, with paired swap-based binomial significance tests (filled bars: FDR < 0.05 under Benjamini–Hochberg correction; unfilled: not significant). (C) Cross-method comparison of three integration tools: Harmony (blue), scVI (orange), and Scanorama (red), on 1,200 cells from each tissue (Pearson scorer, hematopoiesis 76×108 panel). Dashed horizontal line at $\rho = 0$ indicates rank-permutation null.

3.6 Scorer divergence: within-scorer fragility does not transfer across scorer families

To test whether the fragility conclusions drawn under one scorer transfer to another, we scored the hematopoiesis_76$\times$108 panel on a shared 800-cell kidney sample using Pearson correlation, mutual information, GRNBoost2, GENIE3, and scGPT attention, and computed the pairwise agreement (Table 3).

The pattern is clear: the two tree-ensemble scorers (GRNBoost2 and GENIE3) produce nearly-identical rankings (Spearman $\rho = 0.93$, top-500 Jaccard $= 0.54$, RSS $= 0.29$), as expected since both use feature-importance aggregations over similar regressors. Pearson and the tree scorers are moderately correlated ($\rho \in [0.52, 0.56]$, top-100 Jaccard ≈ 0.25–0.30). Mutual information is weakly correlated with Pearson ($\rho = 0.20$) and unrelated to the tree scorers ($\rho \leq 0.21$).

Most importantly, **scGPT attention is uncorrelated with every classical scorer**. Spearman ρ ranges from -0.05 (vs Pearson) to $+0.20$ (vs MI); top-100 Jaccard ranges from 0.02 to 0.06; the RSS composite ranges from 0.69 to 0.76, at or above the rank-permutation null mean of

≈ 0.72. In other words, scGPT attention rankings on the same cells and the same edge universe share fewer than 5% of their top-100 edges with any classical scorer.

This is a first-order finding for the robustness narrative. The paper's Axes 1–5 measure *within-scorer* fragility: how stable is one scorer's ranking under preprocessing perturbation? The scorer-divergence table shows that *across-scorer* agreement is at null levels on the same data (Figure 6). The two fragility axes (within-scorer and across-scorer) are orthogonal: choosing a stable scorer does not guarantee that another stable scorer will agree with it. Our recommendations in Table 5 apply within a given scorer family; an analyst switching scorers must re-audit.

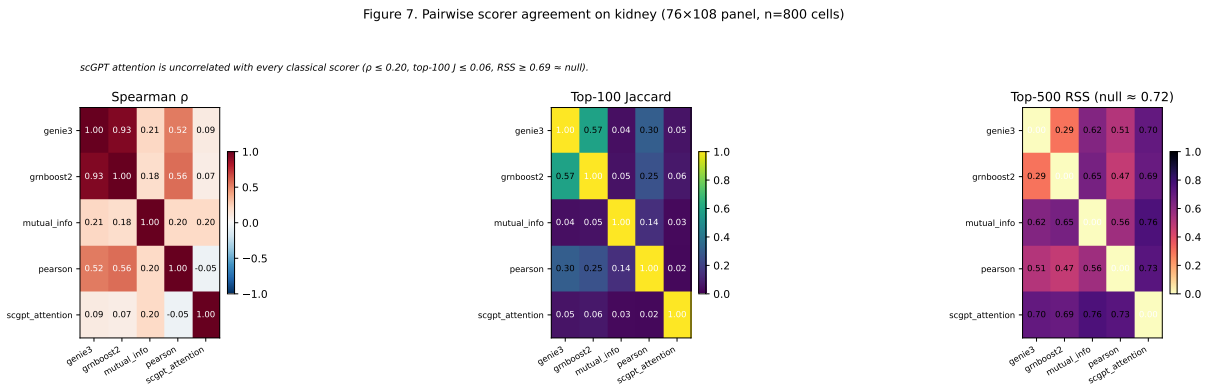

Figure 6: **Pairwise scorer agreement on kidney** (hematopoiesis 76×108 panel, $n = 800$ cells). Left: Spearman ρ. Centre: top-100 Jaccard. Right: top-500 RSS composite (null mean $\approx$ 0.72, shown in darker red). scGPT attention is uncorrelated with every classical scorer (last row/column); GRNBoost2 and GENIE3 are nearly identical.

Table 3: Pairwise scorer-agreement matrix on kidney (76 TFs $\times$ 99 targets panel, $n = 800$ cells). All top-500 RSS values should be compared against the rank-permutation null mean of ≈ 0.72: values far below null indicate concordant rankings, values near null indicate ranking independence.

Scorer A	Scorer B	Spearman	Top-100 J	Top-500 J	Top-500 RSS
Pearson	Mutual info	0.20	0.14	0.21	0.56
Pearson	GRNBoost2	0.56	0.25	0.32	0.47
Pearson	GENIE3	0.52	0.30	0.26	0.51
Pearson	scGPT attention	−0.05	0.02	0.04	**0.73**
Mutual info	GRNBoost2	0.18	0.05	0.14	0.65
Mutual info	GENIE3	0.21	0.04	0.16	0.62
Mutual info	scGPT attention	0.20	0.03	0.06	**0.76**
GRNBoost2	GENIE3	0.93	0.58	0.54	0.29
GRNBoost2	scGPT attention	0.07	0.06	0.06	**0.69**
GENIE3	scGPT attention	0.09	0.05	0.05	**0.70**

3.7 Cross-axis analysis

The five axes were designed independently, but the fragility they reveal is not independent. Table 4 summarizes the per-axis robustness profile across all datasets, and Figure 7 visualizes the pass/marginal/fail status for each axis–dataset combination. Several cross-axis interactions are worth highlighting.

First, the cell-count axis and the rare-type axis are mechanistically linked *under scGPT attention*: because scGPT-attention inference requires >3,000 cells for stability (Axis 1), and most rare cell types contain far fewer than 500 cells, rare types are inherently operating in the unstable regime for scGPT, which provides a partial mechanistic explanation for their 0% reliability-gate passage in Axis 3. For Pearson and tree-ensemble scorers, the MVCC is 200–500 cells, so the direct mechanistic link is weaker; the rare-vs-abundant gap must therefore be driven partly by other factors, likely including expression sparsity and lower null-separation in small cell-type subsamples.

Second, the choice of clustering resolution (Axis 2) affects donor generalization (Axis 4) in a subtle way: coarse clustering pools cell-type variation that would otherwise need to be controlled through per-donor composition matching. This suggests that the predominantly coarse recommendation from dual-null calibration may confer indirect benefits for cross-donor robustness as well.

Third, integration (Axis 5) operates upstream of all other axes by altering the correlation structure on which edge scores are computed. Harmony's rank perturbation therefore has the potential to cascade into cell-count stability, resolution sensitivity, and rare-type null separation, making integration a latent confounder for all other axes.

The most striking finding is that no dataset passes all five axes under their respective calibrated criteria in any single scorer family, and across-scorer agreement is at null levels. Every dataset fails at least two axes, there is no single corrective measure (more cells, coarser clustering, more donors, or integration) that resolves all fragilities simultaneously, and the within-scorer fragility profile does not transfer across scorer families. This reinforces the paper's central message: robustness must be demonstrated, not assumed, along each axis *and* each scorer choice relevant to a given study.

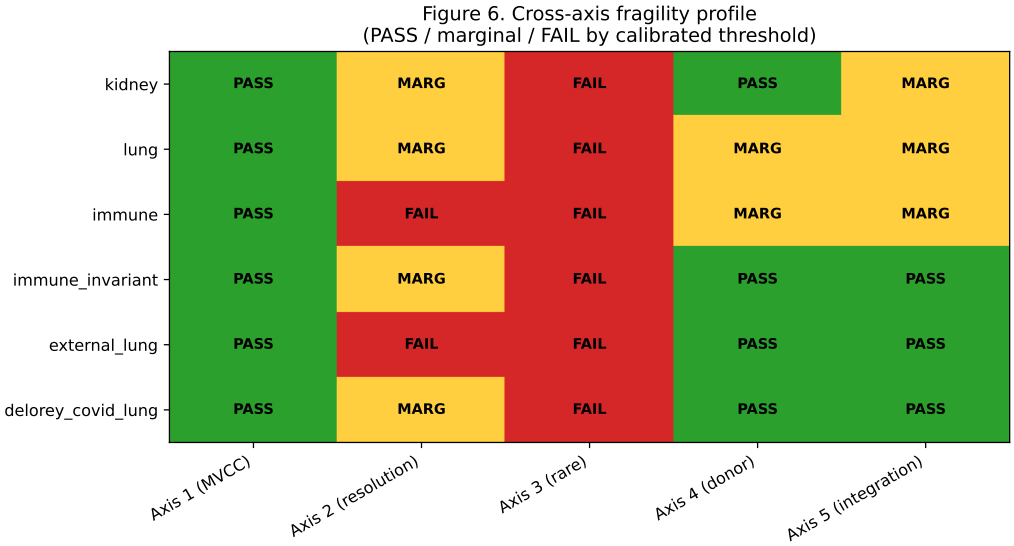

Figure 7: **Cross-axis fragility summary.** Heatmap showing pass (green), marginal (yellow), and fail (red) status for each axis–dataset combination under calibrated criteria.

4 Discussion

This audit demonstrates that GRN edge rankings from single-cell data are fragile along multiple axes simultaneously, and that apparent robustness along one axis (e.g., coarse rank stability for rare types) can mask failure along another (e.g., null separation). The adversarial-review methodology that we applied to each axis is itself a contribution: by requiring corrective experiments before claims are updated, it systematically closes the gap between initial findings and defensible conclusions.

Practical implications. Table 5 provides concrete guidance for practitioners. The central message is that no single parameter choice can guarantee robust GRN inference; instead, robustness must be demonstrated along each relevant axis for the specific tissue, cell count, and analytical choices at hand.

Comparison to existing benchmarks. Published GRN benchmarks (Pratapa et al., 2020; Chen and Mar, 2018) have focused primarily on accuracy relative to gold-standard references, treating preprocessing as fixed. Our audit complements these by asking whether rankings are *stable* across realistic analytical perturbations, a necessary condition for practical utility. The two perspectives are not redundant: a method can be accurate under one configuration and

fragile under perturbation, or robustly mediocre across all configurations.

Limitations. Several limitations apply across the audit. First, edge scoring spans five families (Pearson, mutual information, GRNBoost2, GENIE3, scGPT attention), but none directly estimate causal regulatory effects; although correlation on normalized counts avoids zero-inflation artifacts (Svensson, 2020), it remains susceptible to confounding by shared upstream regulators (Kharchenko et al., 2014). Tree-ensemble scorers partially address confounding but introduce their own biases. scGPT attention reflects patterns learned during pretraining on the CellxGene Census human corpus rather than causal edges per se. Second, the audit covers four Tabula Sapiens tissues, one external-lab lung cohort (Travaglini et al. 2020 SmartSeq2), and one disease cohort (Delorey et al. 2021 COVID-19 lung atlas, 116,313 cells, 27 donors); generalisation to other disease tissues, developmental atlases, or cross-species comparisons remains untested and is a natural next step. Third, the triple-null calibration shows that the *degree-preserving* null is cleared on every cohort, rebutting the concern that constrained shuffles are uniformly too strict; however, the *within-coarse* null is a demanding gate and scoring families with a different null structure might clear it. Fourth, perturbation-backed validation data remain sparse, limiting statistical power for perturbation swap tests. Fifth, three integration methods are reported side-by-side (Harmony (Korsunsky et al., 2019), scVI (Gayoso et al., 2022), Scanorama (Hie et al., 2019)); recent integration benchmarks (Luecken et al., 2022; Tran et al., 2020) note that methods diverge in ways relevant to downstream GRN outputs, and extending our cross-method comparison to additional integration tools is left as future work.

5 Conclusion

We present a systematic multi-scorer multi-axis robustness audit of GRN inference from single-cell transcriptomics, spanning cell count, clustering resolution, rare-cell reliability, donor generalisation, and integration-method sensitivity across five scorer families (Pearson, mutual information, GRNBoost2, GENIE3, scGPT attention) and six cohorts (four Tabula Sapiens tissues, one external-lab SmartSeq2 lung atlas, and one COVID-19 disease atlas). The audit reveals that fragility is pervasive, axis-specific, interactive, and scorer-specific: no dataset passes all five

axes under calibrated criteria in any scorer family, the within-coarse resolution null fails universally under both dual- and triple-null calibrations, the rare-vs-abundant reliability gap is robust to 5,880 threshold combinations, and scGPT attention rankings are statistically uncorrelated with classical-scorer rankings on the same cells.

We provide a unified open-source package ("`fragility`"), consisting of shared preprocessing, panel registry, pluggable scorer engine, null library, metric definitions, and per-axis runners that reproduces every CSV and figure in this paper from raw h5ad via a single `make audit` command. The package is designed to be extensible: adding a new scorer or dataset requires implementing a single interface and registering it in a YAML config. Practitioners can use it to conduct analogous audits for their own datasets, integration pipelines, and scorer families before publishing GRN claims.

Author Contributions

I.K. conceived the study, designed the five-axis robustness audit framework, implemented all computational pipelines, performed all statistical analyses, generated all figures and tables, and wrote the manuscript.

Acknowledgements

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Conflict of Interest

None declared.

Data and Code Availability

The unified `fragility` package (source code, analysis pipelines, null library, metric definitions, panel registry, scorer implementations, and figure-regeneration scripts) is publicly available at https://github.com/Biodyn-AI/grn-robustness-audit and a versioned archive of the code accompanying this submission is deposited on Zenodo at https://doi.org/10.5281/zenodo.20009065. A single `make audit` command reproduces every CSV and figure in this paper from raw h5ad input.

Raw single-cell data are from the following public sources: Tabula Sapiens Consortium (https://tabula-sapiens-portal.ds.czbiohub.org/); Travaglini et al. 2020 SmartSeq2 lung atlas (CellxGene); and Delorey et al. 2021 COVID-19 lung atlas (CellxGene dataset `75c059c8-8fb7-4e6e-a618-a3e01ac42060`). The scGPT *whole-human* checkpoint is from Cui et al. (2024); the DoRothEA and TRRUST panels are from Garcia-Alonso et al. (2019) and Han et al. (2018) respectively.

Supplementary Information

Supplementary Methods S1 contains the panel registry (gene lists for every named TF–target panel used in the paper) and extended preprocessing documentation. Supplementary Methods S2 describes the adversarial AI-assisted review protocol, including base model, prompt rubric, round-by-round logs, and the full-census 27-issue classification under the pre-registered rubric. Supplementary Figures S1–S7 cover, respectively: the full MVCC anchor ablation; the RSS weight-simplex sweep; the rank-permutation empirical null distribution; the 5,880-point reliability-gate threshold-sensitivity heatmap; the top-k scan over $k \in \{100, 500, 1{,}000, 5{,}000, 1\%, 5\%, 10\%\}$; the normalization ablation (depth / size-factor / Pearson residuals); and the triple-null calibration. Supplementary Table S3 reports Axis-3 per-cell-type reliability metrics

under the shared primary panel for all Tabula Sapiens cohorts. All supplementary material is available at *Bioinformatics* online and in the GitHub repository.

Abbreviations

AUC: Area Under the Curve; BH: Benjamini–Hochberg; FDR: False Discovery Rate; GRN: Gene Regulatory Network; MVCC: Minimum Valid Cell Count; PCA: Principal Component Analysis; RSS: Resolution Sensitivity Score; scRNA-seq: Single-Cell RNA Sequencing; TF: Transcription Factor.

References

Aibar, S., González-Blas, C. B., Moerman, T., et al. (2017). SCENIC: single-cell regulatory network inference and clustering. *Nature Methods*, 14(11):1083–1086.

Chen, S. and Mar, J. C. (2018). Evaluating methods of inferring gene regulatory networks highlights their lack of performance for single cell gene expression data. *BMC Bioinformatics*, 19(1):232.

Cui, H., Wang, C., Maan, H., et al. (2024). scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480.

Delorey, T. M., Ziegler, C. G. K., Heimberg, G., et al. (2021). COVID-19 tissue atlases reveal SARS-CoV-2 pathology and cellular targets. *Nature*, 595(7865):107–113.

Dixit, A., Parnas, O., Li, B., et al. (2016). Perturb-Seq: Dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell*, 167(7):1853–1866.

Garcia-Alonso, L., Holland, C. H., Ibrahim, M. M., Turei, D., and Saez-Rodriguez, J. (2019). Benchmark and integration of resources for the estimation of human transcription factor activities. *Genome Research*, 29(8):1363–1375.

Gayoso, A., Lopez, R., Xing, G., et al. (2022). A Python library for probabilistic analysis of single-cell omics data. *Nature Biotechnology*, 40(2):163–166.

Han, H., Cho, J.-W., Lee, S., et al. (2018). TRRUST v2: an expanded reference database of human and mouse transcriptional regulatory interactions. *Nucleic Acids Research*, 46(D1):D199–D202.

Hie, B., Bryson, B., and Berger, B. (2019). Efficient integration of heterogeneous single-cell transcriptomes using Scanorama. *Nature Biotechnology*, 37(6):685–691.

Huynh-Thu, V. A., Irrthum, A., Wehenkel, L., and Geurts, P. (2010). Inferring regulatory networks from expression data using tree-based methods. *PLoS ONE*, 5(9):e12776.

Lause, J., Berens, P., and Kobak, D. (2021). Analytic Pearson residuals for normalization of single-cell RNA-seq UMI data. *Genome Biology*, 22(1):258.

Kharchenko, P. V., Silberstein, L., and Scadden, D. T. (2014). Bayesian approach to single-cell differential expression analysis. *Nature Methods*, 11(7):740–742.

Korsunsky, I., Millard, N., Fan, J., et al. (2019). Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296.

Luecken, M. D. and Theis, F. J. (2019). Current best practices in single-cell RNA-seq analysis: a tutorial. *Molecular Systems Biology*, 15(6):e8746.

Luecken, M. D., Büttner, M., Chaichoompu, K., et al. (2022). Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, 19(1):41–50.

Moerman, T., Aibar Santos, S., González-Blas, C. B., et al. (2019). GRNBoost2 and Arboreto: efficient and scalable inference of gene regulatory networks. *Bioinformatics*, 35(12):2159–2161.

Pratapa, A., Jalihal, A. P., Law, J. N., Bharadwaj, A., and Murali, T. M. (2020). Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nature Methods*, 17(2):147–154.

Svensson, V. (2020). Droplet scRNA-seq is not zero-inflated. *Nature Biotechnology*, 38(2):147–150.

The Tabula Sapiens Consortium. (2022). The Tabula Sapiens: A multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376(6594):eabl4896.

Tran, H. T. N., Ang, K. S., Chevrier, M., et al. (2020). A benchmark of batch-effect correction methods for single-cell RNA sequencing data. *Genome Biology*, 21(1):12.

Wolf, F. A., Angerer, P., and Theis, F. J. (2018). SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19(1):15.

Table 4: Per-axis robustness summary with key metric, calibrated threshold, and practical recommendation.

Axis	**Key Metric**	**Threshold**
Cell count	MVCS (top-1k Jaccard > 0.5)	Scorer-dependent: 200 (Pearson/tree, 4 of 6 cohorts), 500 (Pearson, primary immune and Delorey C
Resolution	Dual-null pass rate	Both nulls must pass
Rare types	Reliability gate (5 criteria)	All must pass
Donor	Fixed-holdout Spearman	90% of best

Table 5: Practical guidelines for robust GRN inference from single-cell data.

Decision Point	Guideline
Sample size	Target $\geq$ 3,000 cells under scGPT attention or use informative prior constraints (DoRothEA / TRRUST) which stabilise scGPT rankings at 200+ cells. Under Pearson, mutual-information, and tree-based scorers, 200–500 cells are often sufficient on the cohorts tested; verify per-scorer MVCC before committing.
Clustering	Default to coarse clustering unless a dual-null calibration (global + within-coarse constrained shuffle) confirms that fine resolution adds separable signal; in our six cohorts, no coarse-vs-fine configuration cleared both nulls, so every recommendation collapses to coarse or hybrid. Report within-class RSS for compartment-level uncertainty.
Rare cell types	Report null-separation AUC alongside rank stability for every cell type; a high Spearman stability can coexist with null-level AUC (Delorey COVID cohort, §3.3). Flag rare types ($<$ 250 cells) as hypothesis-generating. Provide minimum-reliable-size estimates with uncertainty status.
Donor coverage	Collect $\geq$ 6 donors under strict cells-per-donor balancing (300 cells/donor, 8 donors for conservative generalisation); under looser balancing (200 cells/donor), the requirement drops to 2–4 donors. State the cells-per-donor policy explicitly. Use fixed-holdout evaluation, not complementary splits, and report composition-controlled transfer alongside raw transfer.
Integration	Run at least two integration methods on the same cells, not only the tool that happens to be most popular in the pipeline: we observe Spearman ranging from 0.06 (Scanorama on kidney) to 0.87 (Harmony on immune) on identical inputs. Report at least paired swap significance for reference gains, plus one perturbation-backed metric when available, and include a hyperparameter-sensitivity sweep. Avoid universal claims of biological improvement.
Scorer choice	Auditing one scorer does not license conclusions about another: in our kidney experiment, scGPT-attention rankings overlap with Pearson rankings in fewer than 5% of top-100 edges. Re-audit every axis under the scorer used to generate the published rankings.
Cross-axis audit	Report a fragility profile (Table 4) covering all axes relevant to the study design. Acknowledge which axes were not testable and why.